

A Study on Cloud Forensics and Challenges in SaaS Application Environment

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Abstract—With virtualization technique, cloud computing is developed rapidly. However, aspects of the collecting of digital proofs for forensics in cloud computing environment, certain guideline is not yet. Therefore, cloud forensic researches in progress for reactivating security incidents in cloud computing. In this paper, we analyze forensic issue, challenges, already proposed forensic model in previous researches for adopting cloud forensic. And we suggest a novel model and approach for cloud forensic in SaaS and discuss needs of Service Level Agreements (SLA) and possibilities of Forensic-as-a-Service.

Keywords : cloud computing, cloud forensics

I. INTRODUCTION

Profit of enabling network access to a scalable and elastic pool of shareable physical or virtual resources with self-service provisioning and administration on-demand profit, cloud computing market is getting bigger. Because of these characteristics, the number of users using cloud computing is also increased. According to Gartner, they predict the world wide public cloud service market is projected to grow 16.5 percent in 2016 to total \$204 billion, up from \$175 billion in 2015[1]. However, with the growing cloud computing market, security threats began to grow. When a security incident occurs, the most important thing to collect evidence about the incident. While an auditor can access physical machines and collect the digital evidences from traditional ICT system owned by service provider, a cloud service consumer of outsourced cloud computing resources could have very limited rights on the digital evidences needed for incident investigation. This is caused by technical, organizational and legal challenges of cloud computing[2]. For this, the idea of introducing digital forensics in cloud computing have been proposed. According to Ruan[3], The survey received a strong response from the forensic community. Of the 156 survey responses, 75% of the response agreed or strongly agreed that “missing terms and conditions in service level agreement(SLAs) regarding investigations” is a challenge for cloud forensics. In addition, 77% of the respondents agreed or strongly agreed that the “tools provided, techniques supported and access granted regarding forensic investigations should be included in SLAs”. And 83% of the respondents agreed that “a procedure and a set of toolkits in the cloud organization to obtain keys for encrypted data in the cloud” is important or very important[4]. However, the method

using an existing digital forensic there is a difficulty in the cloud as it applies to. Several characteristics of cloud computing complicate the cloud forensic as follows: range of data collection methods, ensuring reliability during data access methods, all process might comply with the law. In this paper, we investigate the issue and the challenges from forensics and current cloud environment, and discuss about cloud forensic in Software-as-a-Service(SaaS) environment considering SaaS maturity model of X.1602 standard[5] and role of cloud computing[6].

The paper is structured as follows. Section 2 review about cloud computing. Section 3 look into issue and process of forensics. Section 4 challenge for forensic introduced in a cloud environment. At this section, find out about the case points to consider, depending on the characteristics of cloud services mentioned in previous chapter. Section 5 discussion about improved layer model in cloud computing and review X.1602 standard for introducing cloud forensics in SaaS environment. Finally, in Section 6 presents conclusions and suggest future research directions.

II. CLOUD COMPUTING

Five essential characteristics of Cloud Computing are on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service[7].

- **On-demand self-service:** cloud service consumer can access different services as needed automatically without provider's intervention.
- **Broad network access:** computing resources are delivered over the network. Consumer can use service when network connection is available environment.
- **Resource pooling:** The resources are pooled to serve the users at a single physical location and/or at different physical location according to the optimality conditions.
- **Rapid elasticity:** The resource can be provisioned without service provider intervention and can be quickly scale in and scale out according to the user needs.
- **Measured service:** Although computing resources are pooled and shared by multiple consumers, the cloud infrastructure is able to use appropriate mechanisms.

According to ISO/IEC 17788, service methods of Cloud Computing are diverse. In cloud community, three main standards are defined. Service, Deploy, Role[8].

First, the cloud community has extensively used the flowing three service models to categories the cloud services.

- **Infrastructure-as-a-Service(IaaS):** cloud service category in which the cloud capabilities type provided to the cloud service customer is an infrastructure capabilities type.
- **Platform-as-a-Service(PaaS):** cloud service category in which the cloud capabilities type provided to the cloud service customer is a platform capabilities type.
- **Software-as-a-Service(SaaS):** cloud service category in which the cloud capabilities type provided to the cloud service customer is an application capabilities type.

There are four cloud deploy models have been in the Cloud community. Public cloud, Private cloud, Community cloud, Hybrid cloud.

- **Public cloud:** cloud services are potentially available to any cloud service customer and resources are controlled by the cloud service provider.
- **Private cloud:** cloud services are used exclusively by a single cloud service customer and resources are controlled by that cloud service customer.
- **Community cloud:** cloud services exclusively support and are shared by a specific collection of cloud service customers who have shared requirements and a relationship with one another, and where resources are controlled by at least one member of this collection.
- **Hybrid cloud:** using at least two different cloud deployment models.

Finally, three roles in cloud computing are defined. Cloud Service Provider(CSP), Cloud Service Customer(CSC), Cloud Service Partner(CSN).

- **Cloud Service Provider(CSP):** party which makes cloud services available
- **Cloud Service Customer(CSC):** party which is in a business relationship for the purpose of using cloud services.
- **Cloud Service Partner(CSN):** which is engaged in support of, or auxiliary to, activities of either the CSP or the CSC, or both

These standards in cloud computing are not new. But these standards are based for apply cloud forensic. There are some examples for understand. 1) in service model phase, CSC only can control and access within their provided service. 2) in deploy model phase, physical access availability in public cloud environment. 3) The range of data collection vary for each role.

III. DIGITAL FORENSICS

A. Digital Forensics

Digital forensic is a branch of forensic science encompassing the recovery and investigation of material found in digital devices, often in relation to computer crime[9]. This system offers functionality similar to an Intrusion Detection System(IDS), but it makes it possible to analyze data from the past. Same as degree of growing cloud computing market, crimes that target cloud computing environment were also increased. So it is issue that suitable forensics application is required for the cloud.

B. Progress of Digital Forensics

The procedure for the forensic investigation are dealing in a variety of research and standards. But the level of detail steps is slightly different. Because the difference in point of view. According to NIST, computer forensics consists of four steps: Collection involves the process of physical acquisition of data. Examination is the process of combing through the data for items of interest. Analysis is the application of the interesting items to the investigative question. Reporting describes the output of analysis[10].

According to Shams Zawoad[11], process of computer forensic consist four steps.

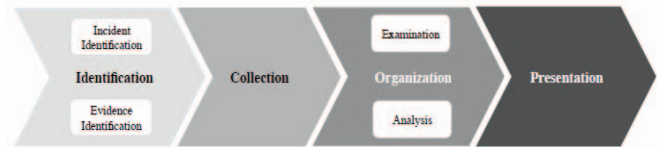


Fig. 1. Process of Computer Forensics[11].

- 1) Identification: To identify that the data collected can be used as evidence.
- 2) Collection: Investigator extracts the digital evidence from variety types of target.
- 3) Organization: Two main steps are required. Examination and analysis of the digital evidence. In examination step, investigator extracts and inspects the data. In analysis step, investigator interprets and correlates the available data to come to a conclusion.
- 4) Presentation: make appropriate report and present to the jury.

The above forensic investigation procedure can be detailed as follows procedures proposed by considering the characteristics of the outsourcing resource, Waldo et al. They suggest another process of computer forensic that add three more steps to define specific method[12]. Because outsourcing resource is one of characteristics in the cloud computing. For the more improve forensic investigation procedure, such as the storage and transportation of data stored in the cloud server is added. Because this is need to guarantee the reliability of such data confidentiality and integrity of data forensic investigation.

- 1) Identification: same as NIST phase.

- 2) Collection: same as NIST phase.
- 3) Transportation: Collected evidence might be transport. Transported area is reliable.
- 4) Storage: While the examination is on-going or even done the evidence needs to be stored in periodically.
- 5) Examination and traces: It consists four sub categories. Analysis, interpretation, Attribution, Reconstruction.
- 6) Presentation: Reporting step. A report containing the outline of the examination proses and the evidence that was found can be created.
- 7) Destruction: The period can range from immediate destruction to seventy years after the case. The end of period, all of related data need to delete.

According to Guo et al, they suggested four steps for cloud forensic process. Until the third process is almost same as the previous one. In the last step, they proposed that investigate the collection of data from three kinds of independent side. This is appropriate approach. Because cloud environment is split into various roles, such as service providers and consumers, partners, etc. So the scope and type of collecting data is different for each role. First, Determine the purpose of the forensics requirement, Identify the types of cloud services, the last process is that determine the type of background technology used. And a three-ways in the final step described below[13].

- Client side: Collect activities performed on client side through live forensics tools. In the case of SaaS, requires the collection of information such as cookies in the browser because it is a web-based service[14]. However, in the most of the CSP does not provide the service or the interface to the data collection. Also it is issue that discovery data and evidence collection for mobile devices[11]. At this time, the list of data that needs to be collected from the Client side is as follows: Intrusion Detection System(IDS), Web Content engine logging, firewalls, access log, chat logs[13].
- Server side: In server side, it can collect log and full evidence from cloud service provider. The list of data that needs to be collected from Server side is access logs, transaction logs, usage logs.
- Developer side: In developer side, it can collect full evidence from developer. And identify the tools used for upload of data. The list of data that needs to be collected from Developer side is firewalls, load balancers, access logs, IDS, NetFlow data.

In order to carry out cloud forensic, the steps above might be incorporated exactly within the cloud environment. However, because there is a variety of different service models (IaaS, PaaS, SaaS), deploy models (Public, Private, Hybrid, Community), etc. there are some difficulties in data collection. Also, within the relationship between CSP and CSC, the SLA is incomplete and most of the CSPs does not provide the service nor the appropriate interface for data collection[11]. In addition, the storage and processing of the data is distributed. It

makes it difficult to collect evidence data. After collecting the data, there are reliability problems with the collected data. For example, when investigating the virtual machine(VM) power off, it is difficult to record volatile data[11]. In order to preserve volatile data in real-time, Brian Hay carried out a study to develop a tool that can monitor volatile data and end the Virtual Machine(VM) once it discovers irregularities without making any changes to a cloud application[15].

Conclusions are as follows. In cloud computing, the more specific and detailed method for introduction of forensic than the current one is needed. So, not only NIST process, likewise Guo's suggestion, dividing the side for successful and accurate data acquisition. Also, it is necessary that the advanced standards in terms of data collection and management might be made for cloud forensics. In the next chapter, to fine out the challenges that make it difficult to solve this problem.

IV. CHALLENGES OF CLOUD FORENSICS

Unlike legacy systems that owns all of the computing resources, in the cloud computing environment, the Cloud Service Provider (CSP) provide infrastructure, platform, application. the CSC utilizes the services provided. This structural difference causes a lot of issues in cloud forensic, such as the storage of data and storage locations, and access the data.

A. Logging challenges

According to Marty et al[16], challenges arising from logging for forensics in cloud environments are as follows.

- 1) Decentralization of logs
- 2) Volatility of logs
- 3) Multiple tiers and layers
- 4) Archival and retention
- 5) Accessibility of logs
- 6) Nonexistence of logs
- 7) Absence of critical information in logs
- 8) Non compatible/ random log formats

The first five challenges can be solved through log management. These are also characteristics of data in cloud environment. The remaining three are more intrinsic problems and have to be addressed through defining logging guidelines and standards. Because these things are only considering about log conditions, a lot of case scenarios can occur and difficult to catch a regular basis. Thus, it might fully consider the standards (Role, location-base, architecture, Service model) in each cloud environment to each use.

B. Cloud Architecture and Role-based approach

In Figure 2, According to Zawoad, cloud environment is divided into six layers: network to access control, as shown in Figure 2 mentioned above. Why difficult to apply forensic in cloud computing is first, data processing is dispersed. Second,

in traditional computer forensics, investigators have full control over the evidence. But it is very hard in cloud environment. Third, there is a lack of reliable evidence as it is difficult to collect evidence due to the multi-tenant features. Fourth, when VM shut down, it is difficult to preserve volatile data. Fifth, chains of custody might clearly depict how the evidence was collected, analyzed, and preserved. Sixth, investigators are completely dependent on CSPs for acquiring evidence[11]. In order to solve the problem, it might be saved for prevent loss of volatile data (e.g. snapshot). And collected evidence might be ensured that the integrity has not been manipulated during the process.

It also shows that CSCs access range in each cloud service. The remaining portions of the layers, it is possible to control the CSP. In SaaS environment, CSP managed application layer to network layer. Because CSC just receives a service through the application from CSP. In PaaS environment, CSC receives support platform that can be developed. So CSC can control up to application layer additionally. Finally, in IaaS environment, due to CSC use the configuration being provide for infrastructure, such as Hardware, network. CSC can control the best large layer than above two service model.

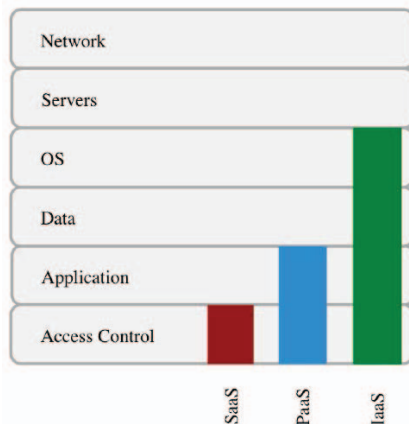


Fig. 2. Customers control over different layers in different service model[11]

In Josiah's paper[17], when collecting evidence with forensic tool, if forensic tool that is installed within the cloud is not reliable. To address these thing, cloud forensic need independent trust model or layer. Acquisition methods and the cumulative trust required in IaaS cloud are given in Table 1.

TABLE I. SIX LAYER OF THE IAAS CLOUD ENVIRONMENT AND POTENTIAL FORENSIC ACQUISITION TECHNIQUES FOR EACH, INCLUDING THE CUMULATIVE TRUST REQUIRED BY EACH LAYER[17]

Layer	Cloud layer	Acquisition method	Trust required
6	Guest application/data	Depends on data	Guest operating system(OS), Hypervisor,

			Host OS, Network
5	Guest OS	Remote forensic software	Guest OS, Hypervisor, Host OS, Hardware, Network
4	Virtualization	Introspection	Hypervisor, Host OS, Hardware, Network
3	Host OS	Access virtual disk	Host OS, Hardware, Network
2	Physical hardware	Access physical disk	Hardware, Network
1	Network	Packet capture	Network

In Table 1, the suggestion has been noted that this might be done a guaranteed reliability of the six-layer forensic data collection. In fact, since the CSP such as Amazon to manage 1 to 2 layer, there are only 3 to 6 layer can control. When examining total system to forensic tools, it has presented successful, but it took a long time to execution. As a result, this shows the necessity to develop the tool to help make the task concise and gather reliable information[17].

C. Service Model-based approach

When divided by the cloud service model based on, challenges for forensic introduced noted in Table 2. In IaaS model, it can be mainly seen that the required reliability in access to the data storage, data management, and distributed environment. In PaaS model, logging can depend on entirely CSP. In SaaS model, same as PaaS, it had a further reduced range that can be controlled it is dependent on the CSP during logging.

TABLE II. CHALLENGES AMONG THE CLOUD SERVICE DELIVERY MODEL[18]

Models	Challenges
Iaas	1. Live forensics and access to volatile data may not be possible (some vendors may not utilize persistent storage) 2. Storage is logical and focused on allocated space; acquisition images may not include data remnants or unallocated disk space 3. Unobtainable failed or obsolete hardware 4. Multi -tenant storage devices may contaminate the acquisition

	5. Logging may be co-located or spread across multiple and changing devices 6. Acquisition may require large amounts of bandwidth to complete in a timely manner 7. Data fragmentation and dispersal 8. Data ownership issues-what happens when the contract is terminated?
SaaS	1. Traditional acquisition is highly unlikely 2. Logging and log details depend on CSP 3. Information may be inconsistent across application programming interfaces (API) 4. Other CSPs may be involved 5. CSP applications may be complex and difficult or impossible to analyze 6. Process/application isolation 7. Systems are more proprietary in nature.
PaaS	1. Logging relies on the CSP environment (system calls may not function in CSP) 2. Systems are more proprietary in nature.

According to ISO, they organized list that potential evidence source under CSP control for cloud forensics. Each service to be provided by the CSP model list are shown in Table 3.

TABLE III. POTENTIAL EVIDENCE SOURCE UNDER CSP CONTROL ABOUT EACH CLOUD SERVICE MODELS[2]

Models	Potential evidence source under CSP control
IaaS	a. Cloud or network provider perimeter network logs b. Logs from DNS servers c. Virtual machine monitor (VMM) logs d. Host operating system logs e. API logs f. Management portal logs g. Packet captures h. Billing records
PaaS	a. Webserver logs b. Application server logs (see SaaS) c. Guest operating system logs d. Host access logs e. Virtualization platform logs f. Network captures g. Billing records h. Management portal logs
SaaS	a. Webserver logs b. Application server logs c. Database logs d. Guest operating system logs e. Host access logs f. Virtualization platform logs and SaaS portal logs g. Network captures h. Billing records

Next, find out the characteristics for reliable collection for cloud forensics.

1) IaaS

In the current IaaS cloud environments not accessible for Volatile data, and store in a position to use multiple physical locations scattered data are due to virtualization technology feels just like in a storage. But there is a challenge of data fragmentation and dispersal exist[13]. It can cause the situation when data may be stored or processed in different jurisdictions on Internet-based cloud computing systems[19].

Since IaaS instances provide much more information that could be used as forensic evidence in case of an incident than the PaaS and SaaS models do, snapshot method is one of the solutions of this. To turn off the VM properly is important, because they contain volatile data. Hence, virtual instances have to be divided into two categories concerning the forensic analysis of the system: shutdown(dead) and running(live). Considering these cases, the huge amount of evidence will collect. So the reliable data storage also needed. Also, the ability to determine how defenses in the virtual environment failed and to what extent the affected systems have been compromised is crucial not only for recovering from an incident. Also forensic investigations gain leverage from such information and contribute to resilience against future attacks on the systems[17].

2) PaaS

According to Birk[20], one of the main advantages of this model is that the core application is under the control of the customer. customer obtains theoretically the power to dictate how the application interacts with other dependencies (databases, storage entities etc.)[17]. In the PaaS services that CSC has a limited range of approaches, logging mechanisms could be implemented which automatically sign and encrypt the log information before its transfer to a central logging server under the control of the customer. Logs might be encrypted and sent. The contents of which can contain the system state, the run time, such as detailed information of the application. For reasons that logs are sent automatically, it is necessary to periodically monitor the outgoing well for the purpose of malicious attacks.

3) SaaS

In the SaaS model, the cloud provider has administrative control over the application layer and total control over the Infrastructure, such as network, servers, operating system. Customer has limited administrative access[3]. In these limited circumstances, it is very difficult to collect evidence themselves. For example, when the application has been investigated, the customer cannot know the position and status of the information needed for the investigation. Therefore, investigator might require a higher level of information to CSP. Moreover, a lot of SaaS CSP like Google offer Single sign-on (SSO) access control to the complete set of their services. Unfortunately, in case of an account compromise, most of the CSP do not offer any possibility for the customer to figure out which data and information has been accessed by the adversary[17]. Therefore, it is necessary to establish the

process for obtaining the forensic information between the client and server.

V. DISCUSSION

We expose and explore various challenge, as mentioned above, to address for cloud forensics. These kind of challenges are from several characteristics of cloud computing and complicate the process of cloud forensics. Digital evidence collection and acquisition, most of all, related with the CSC's control in different layers are key factor to solve these challenges. In addition, up to date, most CSPs do not provide forensic support to their clients via standard APIs or as a part of service level agreement. That's why we are focusing on SLA for cloud forensics and developing SLA component of SaaS application. We, in this section, aim to present our ongoing model, as shown in Figure 3, to address cloud forensics challenge using SLA in cloud SaaS environment and discuss an approach to deduct components of SLA based on the SaaS Maturity model[5], and role of cloud service[6] in SLA perspective.

A. Cloud Forensics SLA Model

In SaaS application environment, refer to the relationship among the CSP, CSN and CSC as shown in Figure 3, CSP and CSN have interfaces with the application, while CSC only interacts with application built upon it. In addition, CSN support CSP as software developer, system integrator, content provider, auditor and so on. The relationship and role impact data collection range and access rights over the data in terms of the service level agreements. For instance, CSC might examine whether CSNs are belong to SaaS application. If so, CSC might understand what role of CSN is and examine that which data is handled and stored by CSN. This causes very practical issue, segregation of duties, forensics acquisition, evidence segregation, etc., for CSC or investigator which have full control over examination.

According to the SaaS maturity model[5], moreover, in SaaS application environment, a SaaS application classified into four levels named as custom, configurable, multi-tenant and scalable level. Each level has different characteristics by independence, sharing and scalability of instance providing the SaaS application. Different maturity levels of the SaaS application need different requirements to SaaS application environments for digital evidence collection and management regarding technical implementation and operational policy. When security incident occurred, for example, isolation of shared instance of multi-tenant level might impact to the other tenant. The custom level has advantage of implementation for their own customized solution for data collection and acquisition than configurable level using common templates.

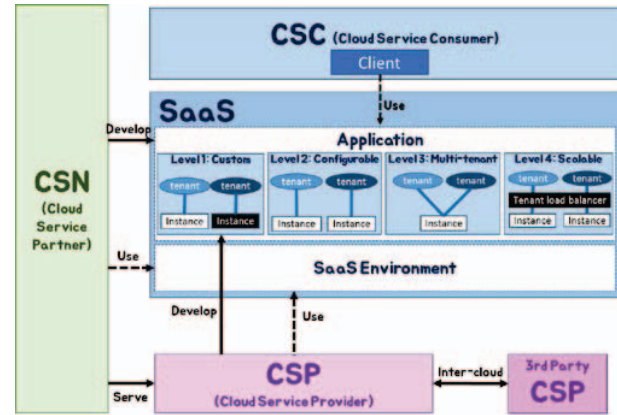


Fig. 3. A Model for Cloud Forensics in SaaS Application Environment

B. Inter-cloud Pattern for CSPs

We might consider inter-cloud pattern such as inter-cloud peering, federation, intermediary as presented in ITU-T Y.3511- Framework of inter-cloud computing[21] between CSP – (3rd Party)CSP. Inter-cloud peering case where two peer CSPs interact with each other, and inter-cloud federation involves using the cloud services within a group of peer CSPs who mutually combine their service capabilities in order to provide the set of cloud services required by CSCs. In the inter-cloud intermediary pattern, the CSP interworks with one or more peer CSPs and provides intermediation, aggregation, and arbitrage of services provided by these CSPs.

These kind of chain of dependency and interoperability involve split of control over data, multi jurisdiction, data ownership in cloud forensics procedure. For SaaS, CSC solely depend on the CSP to get the application log or data. Imagine a CSP provides access control with 3rd party SSO. If a CSC want to be assured access log and history data from a CSP, a CSP needs an agreement with the 3rd party CSP.

VI. CONCLUSIONS

We have demonstrated that describe the cloud for forensic introduced by considering some aspects, and we have been discussing about SaaS maturity model for introducing cloud forensics. For the successful introduction of forensic in cloud computing environment, Many current problems might be resolved by a successful data collection, Training of professional staff, development of appropriate tool, complexity of Chains of custody, cooperation with the CSP, etc[10]. And also consider Presentation, Visualization of Evidence, Cross-Jurisdictional aspects[22]. Especially in SaaS, it might consider all of the four kinds of the above-mentioned maturity models. And it is needed for the proper standards and SLA for each maturity models. In addition, considering the inter-cloud pattern, we can consider the SLA between different roles in a cloud computing environment. Several previous researches were presented Forensic-as-a Service for cloud computing. However, the proposed model didn't describe with specific methods. In future work, we will study about Forensic-as-a-

Service for maturity SaaS model applying the appropriate SLA and standard with inter-cloud pattern.

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